

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	(i) Uses model to form equation with $V = 0$	AO3.4	M1	$\therefore 10 + 100\left(\frac{T}{30}\right)^3 - 50\left(\frac{T}{30}\right)^4 = 0$
	Rearranges to isolate T^4 term	AO1.1a	M1	$\Rightarrow 50\left(\frac{T}{30}\right)^4 = 10 + 100\left(\frac{T}{30}\right)^3$
	Completes rigorous and convincing argument to clearly show the required result. Need to see evidence of division by T to isolate T^3 term Must be an equation throughout AG	AO2.1	R1	$\Rightarrow \frac{T^4}{16200} = 10 + \frac{T^3}{270}$ $\Rightarrow \frac{T^3}{16200} = \frac{10}{T} + \frac{T^2}{270}$ $\Rightarrow T = \sqrt[3]{\frac{162000}{T} + 60T^2}$
(ii)	Calculates T_1 (44.96345.....)	AO1.1a	M1	$T_1 = 44.963$
	Calculates T_2 and T_3 (49.98742....) Condone greater than 3dp (53.50407....)	AO1.1b	A1	$T_2 = 49.987$ $T_3 = 53.504$
(iii)	Explains 38 in context	AO3.2a	B1	38 represents current year 2018

(b)	Translates 2029 into $t = 49$	AO3.3	B1	$10 + 100\left(\frac{t}{30}\right)^3 - 50\left(\frac{t}{30}\right)^4 = 4.5 \times 1.063$
	Uses models to set up equation or evaluate both models at one value of t	AO3.4	M1	$\Rightarrow t = 49.009$ $1980 + 49 = 2029$
	Obtains correct values for both models for two appropriate values of t . $t \in [49, 50]$ eg $t = 49$ and $t = 50$ $t = 49$ gives: 89.89 and 89.81 $t = 50$ gives: 87.16 and 95.47 Or Solves equation using any	AO1.1b	A1	Therefore use of oil and production of oil will be equal in the year 2029

method to obtain AWWF 49.009 to 49.01			
Explains that the use of oil and the production of oil are equal when $t = 49.009$ Or Uses a change of sign argument OE to explain that the value of each model for two appropriate values of t shows that the production of oil and the use of oil are the same for $t \in (49,50)$	AO2.4	E1	
Total 10 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Identifies and clearly defines consistent variables for length and width. Can be shown on diagram.	AO3.1b	B1	Width of rectangle = $2x$ Length of rectangle = $2y$
Models the area of rectangle with an expression of the correct dimensions	AO3.3	M1	$A = 4xy$
Eliminates either variable to form a model for the area in one variable.	AO1.1a	M1	$x^2 + y^2 = 16$
Obtains a correct equation to model the area in one variable	AO1.1b	A1	$A = 4x\sqrt{16 - x^2}$

Differentiates their expression for area. Condone one error	AO3.4	M1	$\frac{dA}{dx} = 4\sqrt{16-x^2} - \frac{4x^2}{\sqrt{16-x^2}}$
Explains that their derivative equals zero for a maximum or stationary point.	AO2.4	E1	$\frac{dA}{dx} = \frac{64-8x^2}{\sqrt{16-x^2}}$ For maximum point $\frac{dA}{dx} = 0$
Equates area derivative to zero and obtains correct value for either variable. CAO	AO1.1b	A1	$\frac{64-8x^2}{\sqrt{16-x^2}} = 0$ $x = 2\sqrt{2}$
Completes a gradient test or uses second derivative of their area function to determine nature of stationary point	AO1.1a	M1	When $x = 2.8$, $\frac{dA}{dx} = 0.448$ When $x = 2.9$, $\frac{dA}{dx} = -1.191$ Therefore maximum
Deduces that the area is a maximum at $x = 2\sqrt{2}$ or $\theta = \frac{\pi}{4}$ Values need not be exact	AO2.2a	R1	The maximum area is 32 sq in
Obtains maximum area with correct units AWRT 32	AO3.2a	B1	
			Total 10 marks

Q3.

EXAM PAPERS PRACTICE

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates rate of change $\frac{dm}{dt}$ into $\frac{dm}{dr}$	AO3.3	M1	$\frac{dm}{dr} = -\frac{k}{\sqrt[3]{m}}$
	Translates inverse proportionality by using $\frac{1}{\sqrt[3]{m}}$ in an equation (no need to see minus sign or k to earn this mark)	AO3.3	M1	
	Forms correct equation with correct notation $\frac{dm}{dr} = -\frac{k}{\sqrt[3]{m}}$ or equivalent	AO1.1b	A1	

	$\text{eg } -\frac{dm}{dt} = \frac{k}{\sqrt[3]{m}} \text{ or } \frac{dm}{dt} = -km^{-\frac{1}{3}}$		
(b)	Gives a relevant criticism of the assumption	AO3.5b	E1 Sam's mass is unlikely to follow this model all the time, when he eats his mass will go up. OR Sam's assumption predicts that his mass will decrease indefinitely.
Total 4 marks			

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies and clearly defines variables.	AO3.1b	B1	Let C = total cost
	Models the cost with an expression of the form $ax^2 + bxh$	AO3.3	M1	x = length of base edges h = length of height $C = 15x^2 + 32xh$
	Eliminates either variable, using volume equation, to form a model for the cost in one variable.	AO3.3	M1	$x^2h = 60$ $h = \frac{60}{x^2}$
	Obtains a correct equation to model cost in one variable	AO3.1b	A1	$C = 15x^2 + \frac{1920}{x}$ (*) Differentiating $30x - \frac{1920}{x^2} = 0$
	Uses their model to find minimum. (at least one term correctly differentiated and equated to zero)	AO3.4	M1	$x^3 = 64$ $x = 4$ $h = \frac{60}{4^2} = 3.75$
	Obtains correct equation	AO1.1b	A1	Height of tank is 3.75 m $\frac{d^2C}{dx^2} = 30 + \frac{3840}{x^3}$
	Obtains correct value for h with correct units in context Award FT from correct substitution into incorrect equation for h but only if all three M1 marks have been	AO3.2a	A1F	$x = 4 \Rightarrow \frac{d^2C}{dx^2} > 0$ therefore minimum ALT from (*) onwards $C = 900h^{-1} + 32\sqrt{60}h^{\frac{1}{2}}$

awarded, must have correct units.			$0 = -900h^{-2} + 16\sqrt{60}h^{-\frac{1}{2}}$
Performs a correct test of 'their' solution: uses the second derivative of 'their' expression for C in terms of x or h to justify that a minimum value for h has been found OE (Second derivative > 0 or test gradient/values either side)	AO2.4	R1	Height of tank is 3.75 m $\frac{d^2C}{dh^2} = 1800h^{-3} - 8\sqrt{60}h^{-\frac{3}{2}}$ $h = 3.75 \Rightarrow \frac{d^2C}{dh^2} = 25.6 > 0$ $\therefore$ minimum

(b)	Explains that two of the sides of the tank will need to have length $x \pm 0.05$ in order to join them	AO3.5c	E1	The sides will need to overlap to be joined, so two of the side lengths will need to be $x + 0.05$
(i)				
(ii)	Explains that the refinement is relatively small and unlikely to have a significant effect on the result.	AO3.5a	R1	The minimum cost is likely to increase slightly, but relative to the size of the tank, an extra 5cm is unlikely to make a significant difference.
Total 10 marks				

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct horizontal component of the initial velocity	AO1.1b	B1	$2.5U = 40$ $U = 16$
	Forms equation to find vertical component of initial velocity	AO3.3	M1	$-10 = 2.5V - 0.5 \times 9.81 \times 2.5^2$
	Obtains correct vertical component of initial velocity	AO1.1b	A1	$V = 8.2625$
	Forms equation for vertical component of velocity at height 3 using 'their' derived values for U and V	AO3.4	M1	$v_y^2 = 8.2625^2 + 2 \times (-9.81) \times 3$
	Obtains correct component of velocity	AO1.1b	A1	$v_y = 3.067...$

	Correct final speed with units, correct for 'their' U and v_y FT applies only if both M1 marks have been awarded	AO3.2a	A1F	$v = \sqrt{16^2 + 3.067^2} = 16.3 \text{ ms}^{-1}$
(b)	States 'their' value of horizontal component of the initial velocity from part (a)	AO3.4	A1F	16 m s^{-1}
(c)	Explains that horizontal velocity has been assumed to be constant in their model and that this is not likely to be true, with valid reasoning	AO3.5b	E1	It was assumed that there were no resistance forces acting on the ball which is unlikely to be true in reality. The horizontal speed of the ball is likely to vary... air resistance would slow the ball down, wind might speed the ball up
Total 8 marks				

Q6.

(a) (i) $t = 0 \quad N = 50$

B1

1

(ii) $t = 24 \quad N = 345$

Must be 345 (not 345.2534..)

B1

1

(iii) $1 + 9e^{\frac{t}{8}} = \frac{500}{400} \Rightarrow 9e^{\frac{t}{8}} = \frac{1}{4}$

Correct algebra seen

M1

$$e^{\frac{t}{8}} = 36$$

$$\text{Or } e^{-\frac{t}{8}} = \frac{1}{36}$$

m1

$$t = 8 \ln 36$$

$$\text{or } t = 16 \ln 6$$

A1

3

(b) (i)
$$\frac{dN}{dt} = -500 \left(1 + 9e^{-\frac{t}{8}} \right)^{-2} \left(-\frac{9}{8} e^{-\frac{t}{8}} \right)$$

Clear attempt at chain rule or quotient rule.

M1

A1

$$= -500 \left(-\frac{1}{8} \left(\frac{500}{N} - 1 \right) \right) \left(\frac{500}{N} \right)^{-2}$$

Use $e^{-\frac{t}{8}} = \frac{1}{9} \left(\frac{500}{N} - 1 \right)$ to eliminate $e^{-\frac{t}{8}}$

m1

$$= \frac{N^2}{500} \left(\frac{1}{8} \left(\frac{500}{N} - 1 \right) \right)$$

$$\frac{dN}{dt} = \frac{N}{4000} (500 - N)$$

Correct algebra to AG

A1

4

Alternatives:

Alternative 1 implicit differentiation

$$e^{-\frac{t}{8}} = \frac{500 - N}{9N}$$

*Correct expressions for $e^{-\frac{t}{8}}$ and attempt to use
attempt to use implicit differentiation*

(M1)

$$\frac{dt}{dN} \left(-\frac{1}{8} e^{-\frac{t}{8}} \right) = -\frac{500}{9N^2}$$

Fully correct

(A1)

use $e^{-\frac{t}{8}} = \frac{1}{9} \left(\frac{500}{N} - 1 \right)$

Attempt to eliminate $e^{-\frac{t}{8}}$ using correct expression

(m1)

to get $\frac{dt}{dN} = \frac{4000}{9N^2} \times \frac{9N}{500-N}$

$$\frac{dN}{dt} = \frac{N}{4000}(500-N)$$

(A1)

(4)

Alternative 2 explicit differentiation

$$t = -8 \ln\left(\frac{500-N}{9N}\right)$$

$$\frac{dt}{dN} = -8 \left(\frac{(500-N)\left(\frac{-1}{9N^2}\right) - \frac{1}{9N}}{\left(\frac{500-N}{9N}\right)} \right)$$

Correct expression for t and attempt at differentiation
with use of chain rule and product for $\ln$ derivative.

(M1)

(A1)

$$= \frac{8}{9N} \left(9 + \frac{9N}{500-N} \right)$$

Clear fractions within fractions

(m1)

$$= \frac{8}{9N} \left(\frac{4500}{500-N} \right)$$

$$\frac{dN}{dt} = \frac{N}{4000}(500-N)$$

All correct

(A1)

(4)

Or

$$t = -8(\ln(500-N) - \ln(9N))$$

$$\frac{dt}{dN} = -8 \left(\frac{-1}{500-N} - \frac{9}{9N} \right)$$

Correct expression for t and $\ln$ derivatives,
condone sign errors

(M1)
(A1)

$$= 8 \left(\frac{1}{500-N} + \frac{1}{N} \right)$$

$$= 8 \left(\frac{N+500-N}{N(500-N)} \right)$$

Common denominator to combine fractions

(m1)

$$= \frac{4000}{N(500-N)} \Rightarrow \frac{dN}{dt} = \frac{4000}{N(500-N)}$$

All correct

(A1)

(4)

Alternative 3 solve differential equation

$$\int \frac{dN}{N(500-N)} = \int \frac{dt}{4000}$$

Separate variables, and attempt to form partial
fractions and integrate to $\ln$ terms $= kt + C$

(M1)

$$\int \frac{1}{500} \left(\frac{1}{N} + \frac{1}{500-N} \right) dN = \int \frac{dt}{4000}$$

(A1)

$$\ln N - \ln(500-N) = \frac{500}{4000}t + C$$

$$(t=0 \ N=50) \ C = \ln \left(\frac{1}{9} \right)$$

Use $(50,0)$ to find C and obtain $e^{\frac{1}{8}t} = f(N)$

(m1)

$$\ln\left(\frac{9N}{500-N}\right) = \frac{1}{8}t \Rightarrow \frac{9N}{500-N} = e^{\frac{1}{8}t}$$

$$N\left(9 + e^{\frac{1}{8}t}\right) = 500e^{\frac{1}{8}t}$$

$$N = \frac{500e^{\frac{1}{8}t}}{9 + e^{\frac{1}{8}t}} = \frac{500}{1 + 9e^{-\frac{1}{8}t}}$$

Manipulate correctly to original given equation.

(A1)

(4)

(ii) $\frac{d}{dN}(500N - N^2) = 500 - 2N$

Differentiate and attempt to find N at max value

M1

$$500 - 2N = 0 \Rightarrow N = 250$$

Condone $\frac{d^2}{dN^2}$ for $\frac{d}{dN}$

A1

$$9e^{\frac{T}{8}} = 1$$

m1

$$e^{\frac{T}{8}} = 9$$

$$T = 8 \ln 9 \quad (17.577)$$

T = 17 or better

CSO

Accept 17, 18, 17.5, 17.6

A1

4

Alternative, by inspection

Max of $N(500 - N)$ occurs at $N = 250$

(B2)

[13]

Q7.

- (a) (i) $3x^2 + 3x^2 + xy + xy + 3xy + 3xy$
correct expression for surface area

M1

$$6x^2 + 8xy = 32 \Rightarrow 3x^2 + 4xy = 16$$

$$2(3x^2 + xy + 3xy) = 32 \text{ etc}$$

AG be convinced

A1

2

- (ii) $(V =) 3x^2 y$ OE
correct volume in terms of x and y

M1

$$= 3x \left(\frac{16 - 3x^2}{4} \right) \text{ or } = 3x^2 \left(\frac{16 - 3x^2}{4x} \right)$$

OE

$$= 12x - \frac{9x^3}{4}$$

CSO AG

be convinced that all working is correct

A1

2

- (b) $\left(\frac{dV}{dx} = \right) 12 - \frac{27}{4} x^2$

one of these terms correct

M1

all correct with 9×3 evaluated (no + c etc)

A1

2

- (c) (i) $x = \frac{4}{3} \Rightarrow \frac{dV}{dx} = 12 - \frac{27}{4} \times \left(\frac{4}{3} \right)^2$
attempt to sub $x = \frac{4}{3}$ into 'their' $\frac{dV}{dx}$

M1

$$\frac{dV}{dx} = 12 - \frac{27}{4} \times \frac{16}{9} = 12 - 12$$

$$\text{or } 12 - \frac{432}{36} = 12 - 12 \text{ or } 12 - \frac{48}{4} = 0 \text{ etc}$$

$$\frac{dV}{dx} = 0 \Rightarrow \text{stationary value}$$

CSO; shown = 0 plus statement

A1

2

(ii) $\frac{d^2V}{dx^2} = -\frac{27x}{2}$ OE

FT for 'their' $\frac{dV}{dx} = a + bx^2$

B1 ✓

when $x = \frac{4}{3}$, $\frac{d^2V}{dx^2} < 0 \Rightarrow \text{maximum}$

or sub of $x = \frac{4}{3}$ into 'their' $\frac{d^2V}{dx^2} \Rightarrow \text{maximum}$

$\left(\text{FT "minimum" if their } \frac{d^2V}{dx^2} > 0 \right)$
EO if numerical error seen

E1 ✓

2

[10]

Q8.

(a) $\left(\frac{dV}{dt} = \right) \frac{3t^2}{4} - 3$

one of these terms correct

M1

all correct (no + c etc)

A1

2

(b) (i) $t = 1 \Rightarrow \frac{dV}{dt} = \frac{3}{4} - 3$

substituting $t = 1$ into their $\frac{dV}{dt}$

M1

$$= -2\frac{1}{4}$$

(- 2.25 OE) BUT must have $\frac{dV}{dt}$ correct

A1cso

2

(ii) Volume is decreasing when $t = 1$

must have used $\frac{dV}{dt}$ in (b)(i) or starts again

because $\frac{dV}{dt} < 0$

must state that $\frac{dV}{dt} < 0$ (or $-2\frac{1}{4} < 0$ etc)

ft increasing plus explanation if their $\frac{dV}{dt} > 0$

E1✓

1

(c) (i) $\left(\frac{dV}{dt} = 0 \Rightarrow\right) \frac{3t^2}{4} - 3 = 0$

PI by "correct" equation being solved

M1

$$\Rightarrow t^2 = 4$$

obtaining $t = k$ correctly from their $\frac{dV}{dt}$

A1✓

$$t = 2$$

withhold if answer left as $t = \pm 2$

A1cso

3

(ii) $\left(\frac{d^2V}{dt^2} =\right) \frac{3t}{2}$

(condone unsimplified) ft their $\frac{dV}{dt}$

B1✓

When $t = 2$, $\frac{d^2V}{dt^2} = 3$ or $\frac{d^2V}{dt^2} > 0$

ft their $\frac{d^2V}{dt^2}$ and value of t from (c)(i)

M1

$\Rightarrow$ minimum

A1cso

3

[11]

Q9.

(a) (i) $\frac{dy}{dt} = \frac{4t^3}{8} - 4t + 4$

one term correct

M1

another term correct

A1

all correct (no + c etc) unsimplified

A1

3

(ii) $\frac{d^2y}{dt^2} = \frac{12t^2}{8} - 4$

ft one term "correct"

M1

correct unsimplified (penalise inclusion of +c once only in question)

A1

2

(b) $t = 2$, $\frac{dy}{dt} = 4 - 8 + 4$

Substitute $t = 2$ into their $\frac{dy}{dt}$

M1

$\frac{dy}{dt} = 0 \Rightarrow$ stationary value

CSO; shown = 0 plus statement

A1

$$t = 2; \frac{d^2y}{dt^2} = 6 - 4 = 2$$

Sub $t = 2$ into their $\frac{d^2y}{dt^2}$

M1

$\Rightarrow y$ has MINIMUM value
CSO

A1

4

(c) (i) $t = 1; \frac{dy}{dt} = \frac{1}{2} - 4 + 4$

Substitute $t = 1$ into their $\frac{dy}{dt}$

M1

$$= \frac{1}{2}$$

OE; CSO

NMS full marks if $\frac{dy}{dt}$ correct

A1

2

(ii) $\frac{dy}{dt} > 0 \Rightarrow$ (depth is) INCREASING

allow decreasing if states that their $\frac{dy}{dt}$

**Reason must be given not just the word
increasing or decreasing**

E1ft

1

[12]

Q10.

(a) (i) S.A. = $4xy + 5xy + 3xy + 6x^2$ OE
Condone one slip or omission

M1

$$= 12xy + 12x^2$$

A1

$$144 = 12xy + 12x^2$$

Must see this line

$$\Rightarrow xy + x^2 = 12$$

AG

A1cso

3

$$(ii) \quad (\text{Volume} =) \frac{1}{2} \times 3x \times 4x \times y \quad \text{OE}$$

M1

$$= 6x^2 \times \frac{(12 - x^2)}{x}$$

$$\text{Must see } (y =) \frac{(12 - x^2)}{x} \text{ or } xy = 12 - x^2$$

for A1

$$(V =) 72x - 6x^3$$

AG must be convinced not working back from answer

A1

2

EXAM PAPERS PRACTICE

$$(b) \quad (i) \quad \frac{dV}{dx} = 72 - 18x^2$$

One term correct

M1

All correct (no + c etc)

A1

2

$$(ii) \quad x = 2 \Rightarrow \frac{dV}{dx} = 72 - 18 \times 2^2$$

$$\Rightarrow \frac{dV}{dx} = 72 - 72 = 0$$

Substitute $x = 2$ into their $\frac{dV}{dx}$

M1

$\Rightarrow$ stationary (value when $x = 2$)
Shown = 0 plus statement
Statement may appear first

A1

2

(c) $\frac{d^2V}{dx^2} = -36x$

FT their $\frac{dV}{dx}$

B1ft

$\frac{d^2V}{dx^2} = -72$ or when $x = 2 \Rightarrow \frac{d^2V}{dx^2} < 0 \Rightarrow$ maximum

*FT their $\frac{d^2V}{dx^2}$ value when $x = 2$
 with appropriate conclusion*

E1ft

2

[11]

Q11.

(a) {Area of sector =} $\frac{1}{2}r^2\theta$
 $\frac{1}{2}r^2\theta$
seen or used for the area

M1

$= \frac{1}{2} \times 8^2 \times 1.4 = 44.8 \text{ m}^2$

Must be exact, not rounded to

A1

2

(b) (i) {Arc =} $r\theta$
 $r\theta$ *seen or used for the arc length*

M1

.... = 11.2

PI Condone AWRT 11.2

A1

Perimeter of sector = $16 + 11.2 = 27.2 \text{ {m}}$

Ft on c's evaluation of 8×1.4

A1F

3

(ii) $27.2 = 2\pi x$

[c's numerical answer for (b)(i)] = $2\pi x$

M1

$$x = \frac{27.2}{2\pi} = 4.329... = 4.33 \text{ to 3sf}$$

Condone > 3 sf

A1

2

[7]

Q12.

(a) (i) $\frac{dx}{dt} = 2t^3 - 40t + 66$

one term correct

M1

another term correct

A1

all correct unsimplified (no + c etc)

A1

3

(ii) $\frac{d^2x}{dt^2} = 6t^2 - 40$

ft one term correct

M1

ft all "correct", 2 terms equivalent

A1ft

2

(b) $\frac{dx}{dt} = 54 - 120 + 66$

substitute $t = 3$ into their $\frac{dx}{dt}$

M1

$= 0 \Rightarrow$ stationary value

CSO

shown = 0 (54 or 2×27 seen) and statement

A1

Substitute $t = 3$ into $\frac{d^2x}{dt^2}$ (= 14)

M1

$\frac{d^2x}{dt^2} > 0 \Rightarrow$ minimum value

CSO; all values (if stated) must be correct

A1

4

(c) Substitute $t = 1$ into their $\frac{dx}{dt}$

must be their $\frac{dx}{dt}$ NOT $\frac{d^2x}{dt^2}$ etc

M1

$\frac{dx}{dt} = 28$

ft their $\frac{dx}{dt}$ when $t = 1$

A1ft

2

(d) Substitute $t = 2$ into their $\frac{dx}{dt}$

must be their $\frac{dx}{dt}$ NOT $\frac{d^2x}{dt^2}$ or x

Interpreting their value of $\frac{dx}{dt}$

M1

$$= 16 - 80 + 66 = 2 \quad (> 0)$$

$\Rightarrow$ increasing when $t = 2$

Allow decreasing if their $\frac{dx}{dt} < 0$

E1ft

2

[13]

Q13.

(a) $V = x(9 - 3x)^2$

Attempt at V in terms of x (condone slip
when rearranging formula for $y = 9 - 3x$)
or $(9 - 3x)^2 = 81 - 54x + 9x^2$

M1

$$\begin{aligned} V &= x(81 - 54x + 9x^2) \\ &= 81x - 54x^2 + 9x^3 \end{aligned}$$

AG; no errors in algebra

A1

2

(b) (i) $\frac{dV}{dx} = 81 - 108x + 27x^2$

One term correct

M1

Another correct

A1

All correct (no + c etc)

A1

$$= 27(x^2 - 4x + 3)$$

CSO; all algebra and differentiation correct

A1

4

(ii) $(x - 3)(x - 1)$ or $(27x - 81)(x - 1)$ etc

"Correct" factors or correct use of formula

M1

$$\Rightarrow x = 1, 3$$

SC: B1, B1 For $x = 1, x = 3$ found by inspection (provided no other values)

A1

2

(c) $\frac{d^2V}{dx^2} = -108 + 54x$ (condone one slip)

ft their $\frac{dV}{dx}$ (may have cancelled 27 etc)

M1

CSO; all differentiation correct

A1

2

(d) (i) $x = 3 \Rightarrow \frac{d^2V}{dx^2} = 54; \quad x = 1 \Rightarrow \frac{d^2V}{dx^2} = -54$

ft their $\frac{d^2V}{dx^2}$ and their **two** x -values

B1ft

1

(ii) ($x =$) 1 (gives maximum value)

Provided their $\frac{d^2V}{dx^2} < 0$

E1

1

(iii) $V_{\max} = 36$
CAO

B1

1

[13]

Q14.

(a) (i) $p \frac{dx}{dt} = q$

where p and q are functions

M1

$$\frac{dx}{dt} = -kx$$

in any correct combination

A1

2

(ii) $-500 = -k \ 20000$ or $500 = k \ 20000$

condone sign error or missing 0

k can be on either side of the equation

M1

$$k = \frac{5}{500} \quad (= 0.025)$$

CSO both (a)(i) and (a)(ii)

A1

2

(b) (i) $A = 1300$

B1

1

(ii) $100 > Ae^{-0.05t}$

condone = for >; condone 99 for 100

M1

$$\ln\left(\frac{100}{A}\right) > -0.05t$$

take logs correctly

condone 0.5

m1

$$t > 51.3$$

or by trial and improvement

(see end of mark scheme)

A1

population first exceeds 1900 in 2059

F if M1 m1 earned and $t > 0$ following A

A1F

4

ALTERNATIVE METHOD

$$t = 51 \rightarrow 101.5$$

$t = 51$ or $t = 52$ considered

M1

$$t = 52 \rightarrow 96.6$$

$$\Rightarrow 51 < t < 52$$

population first exceeds 1900 in 2059

CAO

A3

4

[9]

Q15.

(a) (i) $2x^2 + 2xh + 4xh \quad (= 54)$

Attempt at surface area (one slip)

M1

$$\Rightarrow x^2 + 3xh = 27$$

AG CSO

A1

2

(ii) $h = \frac{27 - x^2}{3x}$ or $h = \frac{9}{x} - \frac{x}{3}$ etc

Any correct form

B1

1

(iii) $V = 2x^2h = 18x - \frac{2x^3}{3}$

AG (watch fudging) condone omission of brackets

B1

1

(b) (i) $\frac{dV}{dx} = 18 - 2x^2$

One term correct "their" V

M1

All correct unsimplified $18 - 6x^2/3$

A1

2

(ii) Sub $x = 3$ into their $\frac{dV}{dx}$

Or attempt to solve their $\frac{dV}{dx} = 0$

M1

Shown to equal 0 plus **statement** that this implies a stationary point if verifying

CSO Condone $x = \pm 3$ or $x = 3$ if solving

A1

2

(c) $\frac{d^2V}{dx^2} = -4x$

(= -12)

FT their $\frac{dV}{dx}$

B1 ✓

$\frac{d^2V}{dx^2} < 0$

at stationary point $\Rightarrow$ maximum

FT their second derivative conclusion

If "their" $\frac{d^2y}{dx^2} > 0 \Rightarrow$ minimum etc

E1 ✓

2

©2025 Exam Papers Practice. All rights reserved.

[10]

Q16.

(a) Sides $24 - 2x$, $9 - 2x$

Either correct

B1

$V = x(24 - 2x)(9 - 2x)$

3 sides involving x multiplied together

M1

$= 4x^3 - 66x^2 + 216x$

AG (be convinced)

A1
3

(b) (i) $\frac{dV}{dx} = 12x^2 - 132x + 216$
Power decreased by 1

M1

One term correct

A1

All correct (no +C etc)

A1
3

(ii) Putting their $\frac{dV}{dx} = 0$ (must see this first)
Or their $12x^2 - 132x + 216 = 0$
Or $12(x^2 - 11x + 18) = 0$ or statement

M1

$\Rightarrow x^2 - 11x + 18 = 0$
AG (be convinced)

A1
2

(iii) $(x - 2)(x - 9) = 0$

M1

Factors, comp sq or formulae used (1 slip)
 $x = 2, x = 9$

A1
2

(iv) Reject $x = 9$, since $9 - 2x < 0$
 $x = 2$ is only possible value

E1
1

(c) (i) $\frac{d^2V}{dx^2} = 24x - 132$

Differentiating their $\frac{dV}{dx}$ (eg $2x - 11$)

M1

Correct

A1

2

(ii) $x=2$ only $\Rightarrow \frac{d^2V}{dx^2} = -84$ (or < 0)

Correct $\frac{d^2V}{dx^2}$ value OE full test.

B1

Maximum value

ft if their test implies minimum

E1 ✓

2

[15]



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

poorly with a surprisingly large minority simply not knowing the correct formula for the circumference of a circle (or an arc-length with angle 2π). The most common wrong answer for those who recalled the correct formula for the circumference was ' $x = 42.7$ ',

obtained by poor use of the calculator in evaluating $\frac{22.7}{2\pi}$ as $\frac{22.7}{2} \times \pi$. Even some of the best candidates lost marks in this opening question while getting high marks elsewhere.

Q12.

In part (a) almost all candidates were able to find the first and second derivative correctly, although there was an occasional arithmetic slip and some could not cope with the fraction term.

Those who substituted $t = 3$ into $\frac{dx}{dt}$ in part (b) did not always explain that $\frac{dx}{dt} = 0$ is the condition for a stationary point. Some assumed that a stationary point occurred when $t = 3$ and went straight to the test for maximum or minimum and only scored half the marks. It

was advisable to use the second derivative test here; those who considered values of $\frac{dx}{dt}$ on either side of $t = 3$ usually reached an incorrect conclusion because of the proximity of another stationary point.

In part (c) the concept of "rate of change" was not understood by many. Approximately

equal numbers of candidates substituted $t = 1$ into the expression for $\frac{dx}{dt}$ or $\frac{d^2x}{dt^2}$ and so only about half of the candidates were able to score any marks on this part. Those who used $\frac{dx}{dt}$ often made careless arithmetic errors when adding three numbers.

In part (d), as in part (c), candidates did not realise which expression to use and many wrongly selected the second derivative. It is a general weakness that candidates do not realise that the sign of the **first** derivative indicates whether a function is increasing or decreasing at a particular point.

Q13.

Some candidates did not attempt part (a). However, those who could square $9 - 3x$ usually earned the marks. A surprising number substituted for both x and y and most were unable to complete the solution successfully.

In part (b)(i), the differentiation was carried out very well by most candidates. Errors in the value of k occurred due to taking out a common factor either before or after differentiating, so values of k such as 3 and 9 were common. A few weak candidates did not recognise that they had to differentiate V here, and generally were unable to proceed through the rest of the question.

In part (b)(ii), the factorisation was usually carried out successfully, and candidates who had not found the correct value of k were still able to earn these marks.

The most common error in part (c) was differentiating $x^2 - 4x + 3$, rather than $27(x^2 - 4x + 3)$, and this approach earned a method mark only.

Credit was given in part (d)(i) for substituting their values of x into their second derivative and so most candidates earned this mark, provided they had two values of x .

In part (d)(ii), most realised that $x = 1$ needed to be selected since $\frac{d^2y}{dx^2}$ was negative when $x = 1$.

In part (d)(iii), not all substituted $x = 1$ into the equation for V . Those who did often made arithmetic errors. Some gave the value of y when $x = 1$ and others gave the answer as

-54 , the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

Q14.

Many candidates seemed to have little experience of formulating a differential equation as often a derivative was missing from their answer to part (a)(i). Attempts at a derivative often involved variables other than the given x and t . Some candidates interpreted the question as 'find x in terms of t ', and many expressions involving an exponential function were seen. Some candidates, who essentially had a correct differential equation, also thought they were required to solve it. Few candidates actually responded to the word 'decreasing' in the question, and many had a minus sign missing from an otherwise correct expression.

The given values of 500 and 20000 were often misinterpreted in part (a)(ii); the 500 often being taken to mean time. However, many candidates did indicate they had understood 500 to be a rate and 20000 the current value of x , but often also had a spurious t or another term in x in their differential equation. Some had a correct value of k , but negative, not apparently noting $k > 0$ had been given in the question. Relatively few fully correct answers to part (a) were seen.

Most candidates found the value of A correctly in part (b)(i), the common error being an apparent misinterpretation of the given information and finding A to be 700.

Part (b)(i) of the question was generally done well with many fully correct answers, and very few solutions by a trial and improvement approach. The majority of candidates could manipulate the expression and take logarithms correctly. Some algebraic errors were seen; in particular those who divided 1900 by 2000, rather than subtracting it, gained no further credit. Some first subtracted the 2000, and then tried to take logarithms of the expression as it stood, usually with an error. Some candidates incorrectly used base 10 logarithms. In completing the question most candidates rounded down their result to give the correct year. Some, who had made an error, did not round down and so had their year incorrect. Some candidates misinterpreted the last request in the question giving their answer as the 51st or 52nd year, rather than 2059.

Q15.

Part (a) Candidates did not seem confident at working on this kind of problem and

algebraic weaknesses were evident. Many worked backwards from the result in part (a)(i) and did not always convince the examiner that they were considering the surface area of four faces and the base. The inability of most candidates to rearrange the formula to make h the subject in part (a)(ii) was alarming. Consequently few, without considerable fudging, could establish the printed formula for the volume. Part (b) Basic differentiation is well

understood and most candidates found $\frac{dV}{dx}$ correctly. Some tried to substitute $x = 3$ into the expression for V in order to show there was a stationary point, but usually this part was answered well.

Part (c) It was not uncommon to see the second derivative as $4x$ even though the first derivative was correct. A generous follow through was given here provided candidates could interpret the value of their second derivative.

Q16.

(a) Once candidates realised that the lengths in centimetres of the sides of the box were $24 - 2x$, $9 - 2x$ and x , they were usually able to multiply out to give the printed answer for the volume. A large number of weaker candidates found this part of the question very difficult and made little or no effort to obtain the printed answer. (a)(i) The expression for dV/dx was usually correct.

(ii) This part was answered too casually by many candidates who simply divided their previous answer by 12 and wrote “= 0” to comply with the printed answer. It was necessary to see that the candidate realised that it is

necessary for $\frac{dV}{dx} = 0$ for stationary points to occur, before dividing their expression for dV/dx by 12.

(iii) Most candidates scored the marks for solving the quadratic equation, although a few who chose to use the formula, rather than factorising, made slips in their arithmetic.

(c) It was pleasing to see that most candidates were able to find a second derivative. However, this was often given as $2x - 11$, since many had previously divided dV/dx by 12 and were using this modified expression. Some candidates did not appear to understand the second derivative test for maxima and minima; they

wrote things such as “ $\frac{d^2V}{dx^2} = 24x - 132 = 0$ when $x = 5.5$ and since this is positive the stationary point is a minimum”.